

Dynamic Programming Volumes I and II

A Quick Introduction — Lecture 1 of 2

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These Lectures

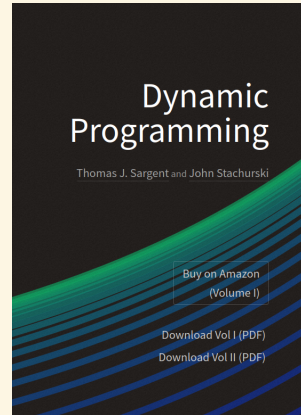
Introduction to DP1 & DP2

Lecture 1: Motivation

1. Review: standard applications / results
2. Variations: how to handle new applications?

Lecture 2: Results

3. Core theory
4. Applications



Background

Dynamic programming (DP) is a **recursive technique for solving optimization problems**

Vast range of applications:

- optimal savings
- real options
- firm investment
- retirement planning
- portfolio allocation

- fiscal policy
- monetary policy
- job search
- dynamic pricing
- capacity allocation
- career choice
- economic geography
- production chains

- natural language processing
- sequential analysis
- automotive cruise control
- hybrid vehicle power management
- L^AT_EX typesetting
- battery consumption
- compiler optimizations
- optimal control of chemical reactors

- protein structure prediction
- RNA folding
- peptide sequencing
- phylogenetic reconstruction
- asteroid mining
- etc., etc.

“NVIDIA’s DPX instructions are built into H100-class GPUs and specifically designed to speed up dynamic programming workloads”

The foundation stone for reinforcement learning (RL)

- Q-learning
- Actor-critic
- GRPO, etc.

Eg. LLM post-training

“DeepSeek-R1 researchers showed that RL can induce emergent reasoning behaviours: self-verification, backtracking, exploring alternative solution paths. The model wasn’t taught these strategies — they emerged from the RL optimization pressure.”

— Claude

RL becoming common in business applications

- Dynamic pricing
- Social media feeds
- Supply chain optimization
- etc.

(Economists' assumption that firms use DP is getting more accurate)

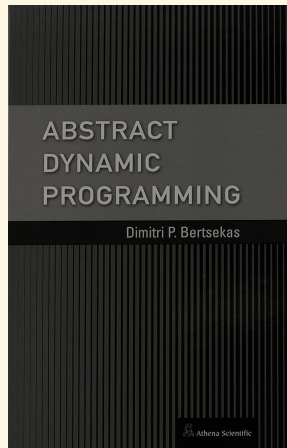
Modern Trends: Variations on a Theme

- recursive preferences
- ambiguity aversion
- adversarial agents / robust control
- nonlinear / negative / state-dependent discounting
- nonstandard representations (e.g., Q-factors, post-action value functions)
- distributional RL
- etc., etc. ...

Can we construct a general theory that covers all these cases?

Existing Literature

- Key references: Bertsekas (2022)
- Works with “generalized Bellman equations”
- Can handle many applications



Steps:

1. Begin with an abstract Bellman equation:

$$v(x) = \max_{a \in \Gamma(x)} B(x, a, v)$$

2. Define optimality of policies

3. Build theory:

- Under what conditions do the usual optimality results hold?
- Which algorithms converge?

DP1–DP2

We construct a more abstract framework

- An **ADP** is a family of maps $\mathbb{T} = \{T_\sigma : \sigma \in \Sigma\}$ on a partially ordered set...
- The **Bellman equation** is $v = \bigvee_{\sigma \in \Sigma} T_\sigma v$
- etc., etc. — explanations in Lecture 2

Now define optimality, provide sufficient conditions, study algorithms

Motivation for the Framework

Motivation 1. Generality

- Capture more applications / variations

Motivation 2. Simplicity

- Proofs of key ideas are “elementary”
- Maximizes effectiveness of machine reasoning

Machine Reasoning – LLMs

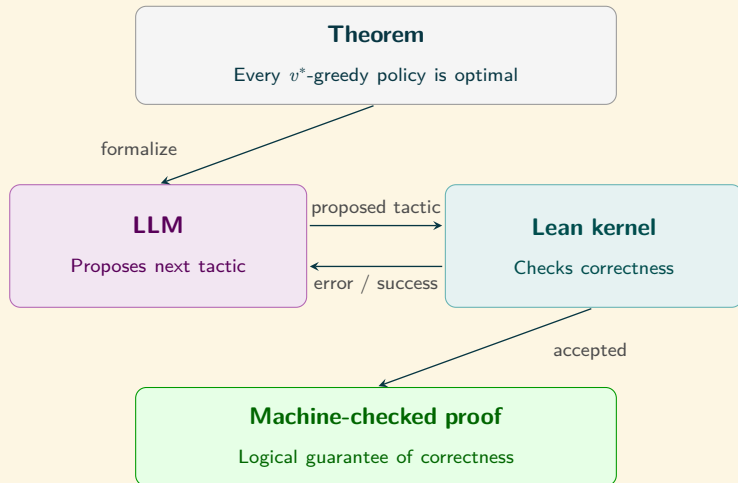
DP1–DP2: An AI-friendly logical framework:

- “1. The reasoning is algebraic, not geometric or topological. This is the kind of thing transformer architectures handle well – pattern matching over structured symbolic expressions.
- 2. The definitions do the heavy lifting. This is a sign of good mathematical design and also makes the proofs easy for AI: there aren’t many “creative” steps, just systematic application of the definitions.
- 3. Your work states every hypothesis, names every property, and never hand-waves. This is exactly the input format AI works best with.”

— Claude

Formalization?

A suitable target for machine-assisted formal reasoning — e.g., Lean?



“The abstract core of DP2 is a near-ideal formalization target – clean algebra on posets, layered proofs, minimal analytic prerequisites.”

“The abstract core is simple enough that formalization is tractable, but rich enough that the interaction between hypotheses is non-obvious. That’s exactly the sweet spot where machine assistance can generate mathematical insight.”

— Claude

Application: Inventory Management

We begin by reviewing a standard inventory problem

Motivation:

- Standardize terminology and notation
- Recall key optimality results
- Recall how the proofs work

Inventory dynamics:

$$X_{t+1} = \max(X_t - D_{t+1}, 0) + A_t$$

The diagram illustrates the inventory dynamics equation $X_{t+1} = \max(X_t - D_{t+1}, 0) + A_t$. It features three arrows pointing to the equation's components: a black arrow from the word "inventory" to X_t , a teal arrow from the word "demand" to D_{t+1} , and an orange arrow from the phrase "order of new stock" to A_t .

Profits:

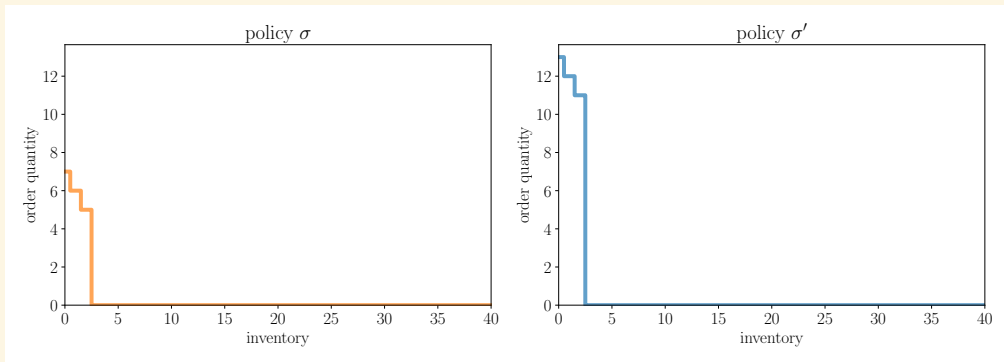
$$\pi_{t+1} = \min(X_t, D_{t+1}) - c A_t - \kappa \mathbb{1}\{A_t > 0\}$$

revenue

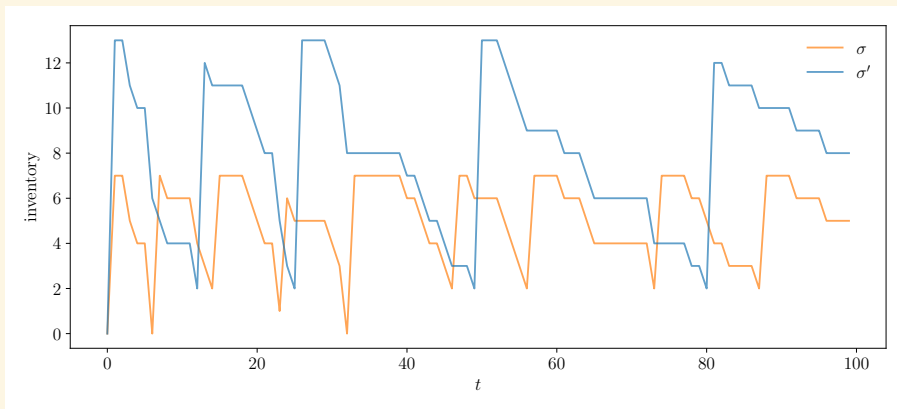
unit cost

fixed cost

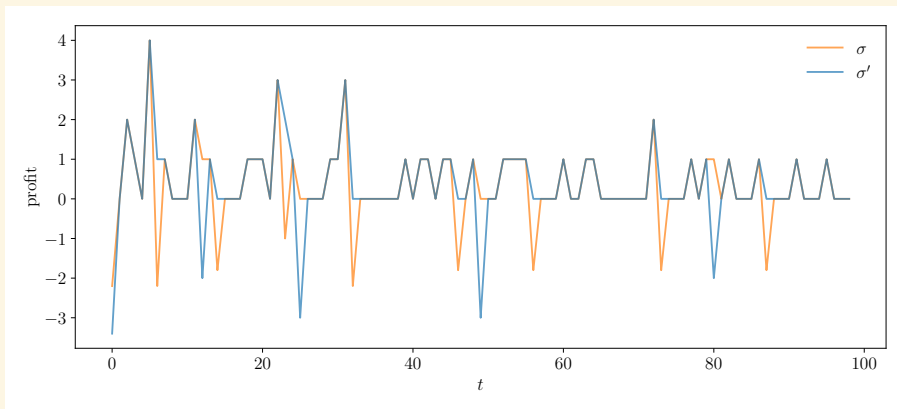
A policy σ maps current inventory to current order size



Inventory dynamics $(X_t^\sigma)_{t \geq 0}$ and $(X_t^{\sigma'})_{t \geq 0}$ under σ and σ'



Profit sequences $(\pi_t^\sigma)_{t \geq 0}$ and $(\pi_t^{\sigma'})_{t \geq 0}$ under σ and σ'



Present value:

$$\sum_{t \geq 0} \beta^t \pi_{t+1}^\sigma$$

Expected present value:

$$v_\sigma(x) := \mathbb{E} \left[\sum_{t \geq 0} \beta^t \pi_{t+1}^\sigma \right]$$

(EPV of σ conditional on initial inventory x)

EPV recursion:

expected present value under σ
given current state

$$v_{\sigma}(x) = \mathbb{E}[\pi_{\sigma}(x, D)] + \beta \mathbb{E}[v_{\sigma}(h_{\sigma}(x, D))]$$

current profits
under σ

next period
inventory under σ

The diagram illustrates the components of the EPV recursion equation. A blue arrow points from the text 'expected present value under σ given current state' to the left-hand side of the equation, $v_{\sigma}(x)$. Another blue arrow points from the same text to the right-hand side of the equation, specifically to the term $v_{\sigma}(h_{\sigma}(x, D))$. A red arrow points from the text 'current profits under σ ' to the term $\mathbb{E}[\pi_{\sigma}(x, D)]$. A green arrow points from the text 'next period inventory under σ ' to the term $v_{\sigma}(h_{\sigma}(x, D))$.

Equivalently, v_σ is the fixed point of the **policy operator**

$$(T_\sigma v)(x) = \mathbb{E} [\pi_\sigma(x, D)] + \beta \mathbb{E} [v(h_\sigma(x, D))]$$

Properties

- A **contraction** map with unique fixed point $v_\sigma \in V := \mathbb{R}^X$
- **Order-preserving**: $v \leq w \implies T_\sigma v \leq T_\sigma w$

Computation: Fix v and iterate to get $\lim_{k \rightarrow \infty} T_\sigma^k v = v_\sigma$

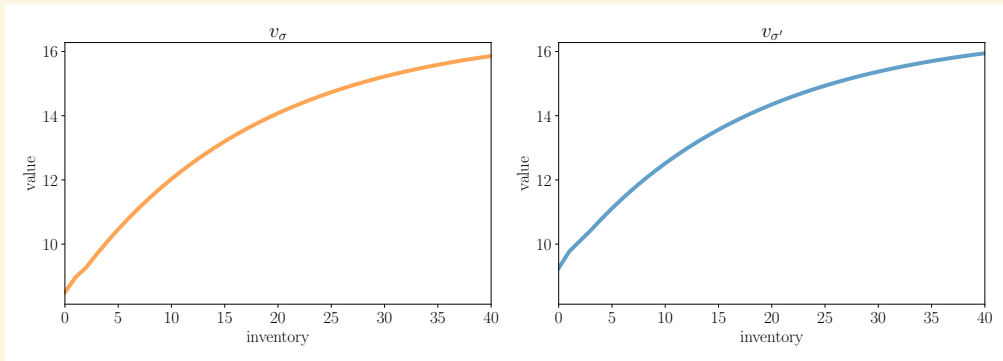


Figure 1: Lifetime value functions v_σ and $v_{\sigma'}$

Value function:

$$v^*(x) := \sup_{\sigma \in \Sigma} v_{\sigma}(x)$$

A policy $\sigma \in \Sigma$ is called **optimal** if

$$v_{\sigma} = v^*$$

- Does such a policy exist?
- If so, how should we compute it?

The **Bellman operator** is

$$(Tv)(x) = \max_{a \in \Gamma(x)} \{ \mathbb{E}[\pi(x, a, D)] + \beta \mathbb{E}[v(h(x, a, D))] \}$$

Function v satisfies the **Bellman equation** if $Tv = v$

Given any v , a policy σ is called **v -greedy** if

$$\sigma(x) \in \arg \max_{a \in \Gamma(x)} \{ \mathbb{E}[\pi(x, a, D)] + \beta \mathbb{E}[v(h(x, a, D))] \} \quad \forall x \in X$$

Note: This is **equivalent** to $T_\sigma v = Tv$

Optimality

The following statements are true

1. The value function v^* is the unique solution to the Bellman equation in V
2. A policy σ is optimal if and only if it is v^* -greedy
3. At least one optimal policy exists

Below we will say that the “fundamental optimality properties hold”

Proof: due to Bellman, Blackwell — see, e.g., [Stokey and Lucas \(1989\)](#) or [Puterman \(2005\)](#)

Proof

The following facts are used:

- Each T_σ is order preserving ($v \leq v'$ implies $T_\sigma v \leq T_\sigma v'$)
- Each T_σ is a contraction map
- T is order preserving ($v \leq v'$ implies $T v \leq T v'$)
- T is a contraction map

Algorithms

The big three:

1. Value function iteration (VFI)
2. Howard policy iteration (HPI)
3. Optimistic policy iteration (OPI)

There are countless variations on these ideas

- Eg. actor-critic, core method for RL, is a variation on OPI

Algorithm 1: Value function iteration (VFI)

input $v \in V$, an initial guess of v^*

input tol , a tolerance level for error

$\varepsilon \leftarrow \text{tol} + 1$

while $\varepsilon > \text{tol}$ **do**

$v' \leftarrow Tv$

$\varepsilon \leftarrow \|v' - v\|$

$v \leftarrow v'$

end

return a v -greedy policy σ

Algorithm 2: Howard policy iteration (HPI)

input $\sigma \in \Sigma$ and tolerance tol

set $v \leftarrow$ the lifetime value v_σ

set $\varepsilon \leftarrow \text{tol} + 1$

while $\varepsilon > \text{tol}$ **do**

$\sigma \leftarrow$ a v -greedy policy

$v' \leftarrow$ the lifetime value v_σ

$\varepsilon \leftarrow \|v' - v\|$

$v \leftarrow v'$

end

return σ

Algorithm 3: Optimistic policy iteration (OPI)

input $m \in \mathbb{N}$ and tolerance tol

input $v \in V$, an initial guess of v^*

$\varepsilon \leftarrow \text{tol} + 1$

while $\varepsilon > \text{tol}$ **do**

$\sigma \leftarrow$ a v -greedy policy

$v' \leftarrow T_{\sigma}^m v$

$\varepsilon \leftarrow \|v' - v\|$

$v \leftarrow v'$

end

return σ

Loosely speaking

- VFI is fixed point iteration on the Bellman operator
- HPI is Newton iteration on the Bellman operator
- OPI is a convex combination of the two

For the inventory problem we have the following results:

Theorem 1: Convergence of Algorithms

The following statements are true

1. The sequence of iterates $(v_k^{\text{VFI}})_{k \geq 0}$ converges to v^*
2. The sequence of iterates $(v_k^{\text{OPI}})_{k \geq 0}$ converges to v^*
3. The sequence of iterates $(v_k^{\text{HPI}})_{k \geq 0}$ converges to v^*

In addition, HPI returns an exact optimal policy in finitely many steps

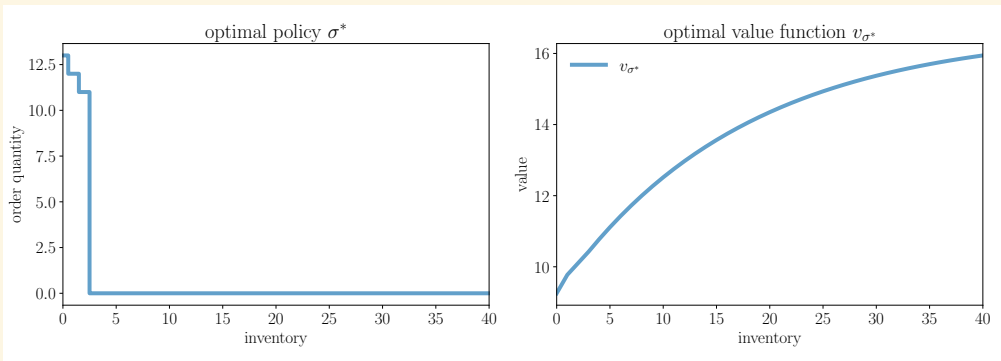


Figure 2: Optimal policy and value function

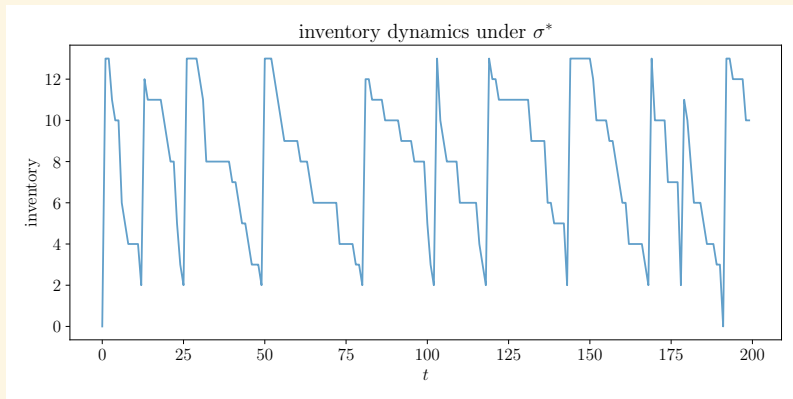


Figure 3: Inventory dynamics under the optimal policy

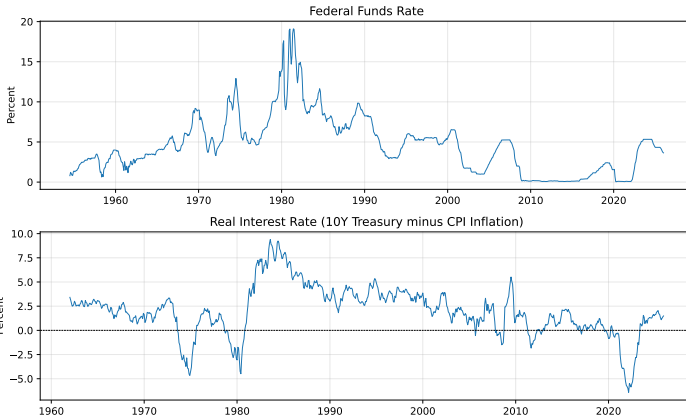
Extensions

Until now we have

- studied a traditional problem
- recalled standard results
- reviewed how the proofs work (operators are order preserving contractions)
- discussed the three major algorithms

Now let's look at some variations

Extension 1: State-Dependent Discounting



The new Bellman equation is

$$v(x, z) = \max_{a \in \Gamma(x)} \mathbb{E} \left[\pi(x, a, D) + \beta(z) \int v(h(x, a, D), z') Q(z, dz') \right]$$

Theoretical issues

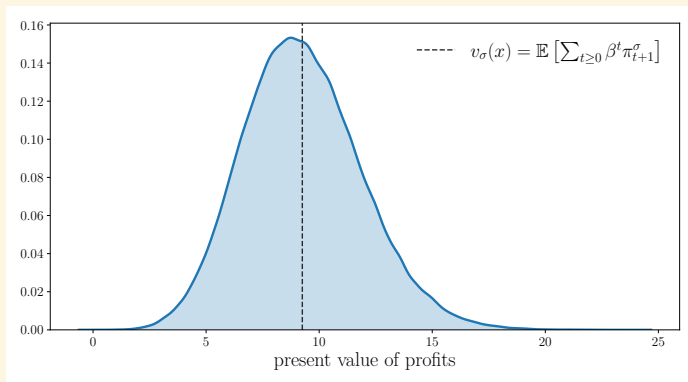
- T_σ and T are nonstandard
- and not always contractions

Do the fundamental optimality properties hold? Do the algorithms converge?

Extension 2: Handling Risk

Optimal policy σ^* solves

$$\max_{\sigma \in \Sigma} v_{\sigma}(x) = \max_{\sigma \in \Sigma} \mathbb{E} \left[\sum_{t \geq 0} \beta^t \pi_{t+1}^{\sigma} \right]$$



Distribution of $\sum_{t \geq 0} \beta^t \pi_{t+1}^\sigma$:

Risk neutrality is the benchmark with complete markets / full information

- Modigliani and Miller (1958)
- Arrow (1964)
- Diamond (1967)

But in practice many firm decisions are not risk neutral

- Tufano (1996)
- Coles et al. (2006)
- Bodnar et al. (2019)

“When we tested how 1,500 executives from 90 countries reacted to different investment scenarios, we discovered that they demonstrated high levels of risk aversion regardless of the size of the investment.”

— Koller et al., [Overcoming a Bias Against Risk](#) (McKinsey, 2017)

“Managers acknowledged that although their risk aversion was bad for their companies, it was good for their careers.”

— Rudd & Lawson, [Your Company Is Too Risk-Averse](#) (HBR, 2020)

To handle risk perhaps we should solve

$$\max_{\sigma \in \Sigma} \mathcal{E} \left[\sum_{t \geq 0} \beta^t \pi_{t+1}^\sigma \right] \quad \text{for suitable nonlinear } \mathcal{E}$$

Eg. Mean-variance criterion

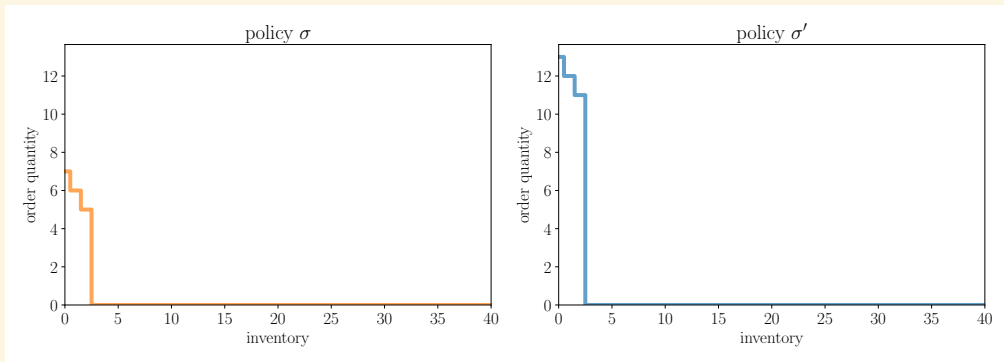
$$\mathcal{E}[X] = \mathbb{E}[X] - \frac{\gamma}{2} \text{Var}[X]$$

Eg. Mean minus value-at-risk

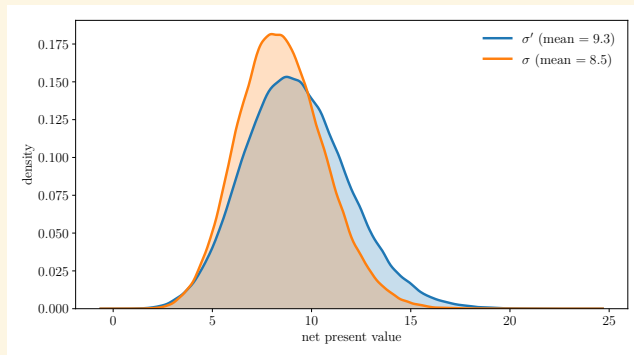
$$\mathcal{E}[X] = \mathbb{E}[X] - \frac{\gamma}{2} \text{VaR}_{\alpha}[X]$$

Now which policy is best?

Recall these two policies:

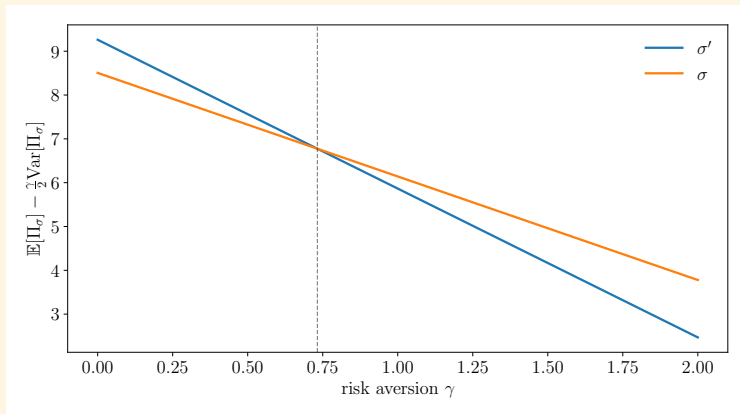


Distributions of $\sum_{t \geq 0} \beta^t \pi_{t+1}^\sigma$ and $\sum_{t \geq 0} \beta^t \pi_{t+1}^{\sigma'}$



Policy σ' has higher mean but more spread

Under mean-variance, for large γ , the conservative policy σ is preferred:



Can we **use DP theory** to solve

$$\max_{\sigma \in \Sigma} \mathcal{E} \left[\sum_{t \geq 0} \beta^t \pi_{t+1}^{\sigma} \right]$$

- In general we can't!
- Even the question is wrong: nonstationary policies can do better
- And this leads to time inconsistency. . .

Researchers are trying to fix these problems with varying degrees of success

- A nice example is [Bäuerle and Jaśkiewicz \(2025\)](#)

Back to Recursion

Workaround: apply $\mathcal{E}$ **period by period**

Eg. Lifetime value of a policy is a fixed point v_σ of

$$(T_\sigma v)(x) = \mathcal{E} [\pi_\sigma(x, D) + \beta v(h_\sigma(x, D))]$$

Here

- $\mathcal{E}$ is a possibly nonlinear “risk-adjusted expectation”
- T_σ is a new **policy operator**

Valuations are recursive **by construction**

Recursive...but traditional theory does not cover this case

- T_σ and T are nonstandard
- and not always contractions

Do the fundamental optimality properties hold?

Do the big three algorithms converge?

Some Additional DP Problems

We want a general theory of DP that can cover the preceding cases

But wait!

There are still more interesting DP problems

Let's list some more DP problems to get a more complete accounting of what we need

Example: Sequential Analysis (Wald)

Hidden state is θ_0 or θ_1 and π = current belief that $\theta = \theta_1$

The Bellman equation:

$$\ell(\pi) = \min \left\{ \pi L_0, (1 - \pi) L_1, c + \int \ell(\pi') P(\pi, d\pi') \right\}$$

- T_σ and T are order preserving — but not contracting
- Do the fundamental optimality properties hold?
- How about convergence of algorithms?

Example: Stochastic Shortest Path

The Bellman equation is

$$\ell(x) = \min_{a \in \Gamma(x)} \left\{ c(x, a) + \sum_{x'} \ell(x') P(x, a, x') \right\}$$

- T_σ and T are order preserving — but not contracting
- Do the fundamental optimality properties hold?
- How about convergence of algorithms?

Example: Epstein–Zin

The Bellman equation is

$$v(w) = \max_{0 \leq c \leq w} \left[(1 - \beta) c^{1-1/\psi} + \beta \left(\int v(R(w - c) + y)^{1-\gamma} \varphi(dy) \right)^{\frac{1-1/\psi}{1-\gamma}} \right]^{\frac{1}{1-1/\psi}}$$

- T_σ and T are order preserving — but not contracting in general
- Do the fundamental optimality properties hold?
- How about convergence of algorithms?

Example: Irreversible Investment with Ambiguity

Manager fears misspecification

Chooses to maximize against worst-case alternatives

$$v(k, z) = \max_{0 \leq i} \left\{ f(k, z) - i + \beta \inf_{\psi \ll \varphi} \left[\int v(k', g(z, x)) \psi(dx) + \frac{1}{\gamma} D(\psi \parallel \varphi) \right] \right\}$$

- T_σ and T are order preserving — but not contracting in general
- Do the fundamental optimality properties hold?
- How about convergence of algorithms?

Example: Nonlinear Discounting

Job search with nonlinear discounting:

$$v(w) = \max \left\{ e(w), c + \int \beta[v(w')]P(w, dw') \right\}$$

- allows discounting to depend on continuation values
- can model “magnitude effects” — large rewards discounted less
- T_σ and T are order preserving — but not contracting in general
- Do the fundamental optimality properties hold?
- How about convergence of algorithms?

Negative Discount Rates

Consider a DP problem with Bellman equation

$$v(x) = \min_{0 \leq a \leq x} \{c(a) + \delta v(x - a)\}$$

- c measures cost of effort with $c' > 0$ and $c'' > 0$
- $\delta > 1$ for negative discount rates — want to get task “over and done with”
- T_σ and T are order preserving — but not contracting
- Do the fundamental optimality properties hold?
- How about convergence of algorithms?

Example: Structural Estimation

Model agents as solving DPs; infer parameters θ to match model outputs to data

Post-action value function g satisfies a nonstandard Bellman equation:

$$g(x, a) = \int \max_{a' \in A} [r(x', a') + \beta g(x', a')] P(x, a, dx')$$

- Integration and maximization are **reversed** relative to standard DP
- g depends on both state x and action a
- Doesn't fit classical DP framework or Bertsekas (2022) — how can we handle it?

Example: Distributional DP/RL

Setup: Finite MDP with Bellman equation

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{x'} v(x') P(x, a, x') \right\}$$

Traditional DP: maximize $v_{\sigma}(x) = \mathbb{E} [Z_{\sigma}(x)]$ where

$$Z_{\sigma}(x) = \sum_{t \geq 0} \beta^t r(X_t, \sigma(X_t))$$

A **distributional value function** is a map $x \mapsto \eta(x, \cdot)$

Assigns a return distribution to each state

- Tracks an estimate of $\eta_\sigma(x, \cdot) := \text{distribution of } Z_\sigma(x)$
- Captures variance, quantiles, tail risk
- Core idea behind distributional RL (Bellemare et al. 2017)

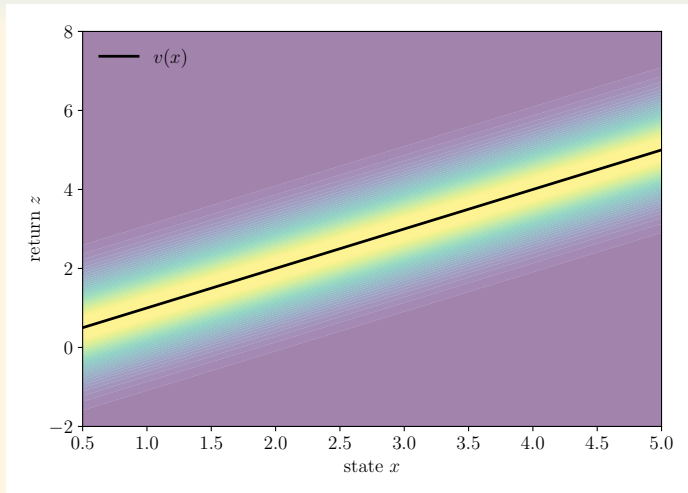


Figure 4: Distributional value function $\eta(x, \cdot)$ and scalar value $v(x)$

The **distributional policy operator** D_σ maps distributional value functions to distributional value functions:

$$(D_\sigma \eta)(x, B) = \int \left[\int \mathbb{1}_B(r_\sigma(x) + \beta v) \eta(x', dv) \right] P_\sigma(x, dx')$$

Key properties:

- D_σ is **order preserving** under pointwise stochastic dominance
- D_σ is a **β -contraction** in the supremum Wasserstein-1 metric
- The unique fixed point $\eta_\sigma(x, \cdot)$ gives the distribution of $Z_\sigma(x)$ at each x

The scalar value function is recovered: $v_\sigma(x) = \int z \eta_\sigma(x, dz)$

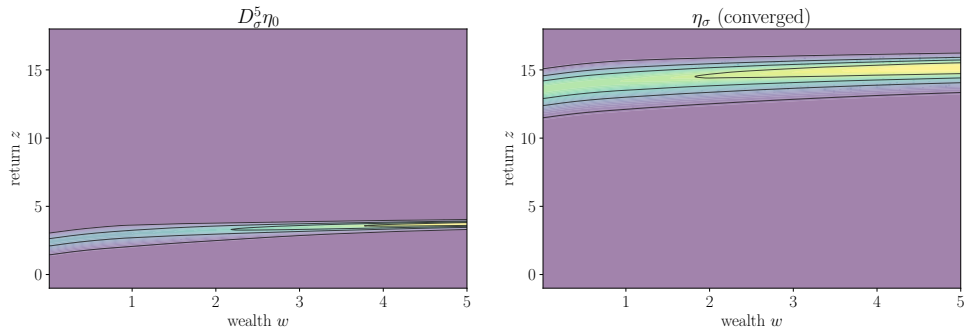


Figure 5: Iterating D_σ : early iterate and converged η_σ

Questions:

- Can we build an abstract theory general enough to handle both distributional and traditional DP?
- If so, what can we learn about distributional DP?

State of Play

The set of possible applications is extremely broad

Traditional assumptions often fail:

- failure of contractivity
- failure of additive separability
- nonstandard timing
- nonstandard value spaces

and some are preserved

- recursivity, order-preserving T and T_σ

DP1–DP2 build an abstract DP framework that can handle these cases

- DP1 considers the case of finite states and actions — cleanest
- DP2 considers the general case

Lecture 2 gives an introduction to the key ideas

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